

L05: Support Vector Machines

CSci 574 Machine Learning (Géron) Ch. 5

Derek Harter

Department of Computer Science
East Texas A&M University

Summer 2025

Linear SVM Classification

- When two classes can be separated by a line (hyperplane). Figure 1, left
 - They are **linearly separable**
- SVM classifier fits the widest possible street. Figure 1, right
 - This is called **large margin classification**

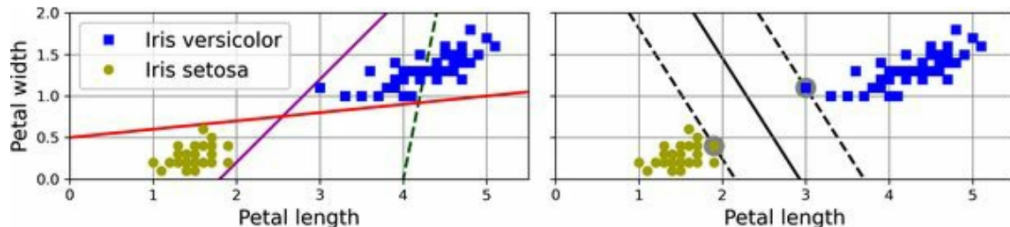


Figure 1: Large margin classification (Géron, 2023, pg.287)

Géron, A. (2023). *Hands-on machine learning with scikit-learn, keras and tensorflow* (third).
O'Reilly Media, Inc.